

TOWARDS NON-ABELIAN LOCALIZED BOST–CONNES SYSTEMS: THE MATRIX KAKUTANI DICHOTOMY AND A TANNAKIAN OBSTRUCTION

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ABSTRACT. The GL_1 theory of [5] classifies the KMS_β simplex of $\mathcal{A}_{K,S}$ as $\text{Prob}(G_S/\Xi_\beta^\perp)$, where Ξ_β is a subgroup of the Pontryagin dual $\widehat{G_S}$ cut out by a Dirichlet series condition. We examine what survives when the abelian symmetry group is replaced by a Galois group with non-abelian image, and separate three things that are usually conflated.

What transfers. The Dirichlet-series half of the Kakutani dichotomy is representation-theoretic, not character-theoretic. For a unitary representation ρ of a compact group equipped with Frobenius classes we take as the obstruction quantity the Hilbert–Schmidt Frobenius series

$$\Xi_\beta(\rho, S) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} \|I_n - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2 = 2 \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} (n - \Re \text{Tr } \rho(\text{Frob}_{\mathfrak{p}})),$$

which reduces to the abelian condition for characters, and whose convergence is the coboundary condition for the $U(n)$ -valued Frobenius cocycle whenever that cocycle is defined, namely when the Frobenius classes commute; without commutation there is no cocycle on the tail relation at all, and the series is the primary object. Its relation to L -functions is

$$\Xi_\beta(\rho, S) = 2n \log \zeta_{K,S}(\beta) - 2\Re \log L_S(\beta, \rho) + O(1),$$

so that at $\beta \leq 1$ and $S = P_K$ the obstruction is absent for every nontrivial irreducible ρ , the input being $L(1, \rho) \neq 0, \infty$: Hecke–Landau for Artin representations, Jacquet–Shalika in the automorphic case. For Artin representations the Chebotarev criterion of [5, Thm. 3.9] transfers with $D_\rho = \{\mathfrak{p} : \rho(\text{Frob}_{\mathfrak{p}}) \neq I\}$.

What does not transfer, and its replacement. The classification $\text{Prob}(G_S/\Xi_\beta^\perp)$ uses Pontryagin duality twice and fails without it: for non-abelian G the set of irreducible representations is not a group, Ξ_β cannot be a subgroup of it, and $\Xi_\beta^\perp$ is undefined. The correct replacement is not a subgroup of a dual but a *quotient of the group*. We prove that $\mathcal{T}_\beta(S) = \{\rho : \Xi_\beta(\rho, S) < \infty\}$ is closed under direct sums, subobjects, tensor products and duals, hence is a Tannakian subcategory of $\text{Rep}(G)$, and therefore equals $\text{Rep}(G/N_\beta)$ for a closed normal subgroup $N_\beta = \bigcap_{\rho \in \mathcal{T}_\beta} \ker \rho$. In the abelian case $N_\beta = \Xi_\beta^\perp$ and the two formulations agree.

What is not proved here. We do not construct a GL_n algebra whose KMS_β simplex is $\text{Prob}(G/N_\beta)$, and we explain precisely which step of the GL_1 proof has no available substitute. Two concrete findings bear on the GL_2 case, formulated for the Sato–Tate classes of a non-CM elliptic curve. If the classes along S are Sato–Tate equidistributed then $\Xi_\beta(\rho, S) \sim 4 \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta}$ diverges whenever the partition function does, so *no* symmetry breaking arises from a Sato–Tate representation. Conversely, assuming Sato–Tate as an equidistribution input, we construct sparse S with $\zeta_S(1) = \infty$ and $\Xi_1(\rho, S) < \infty$ for the representation attached to a non-CM elliptic curve, by selecting blocks of primes with $\theta_p \leq 2^{-j}$; the exotic phase is therefore available, but only on deliberately constructed anomalous sets, never generically.

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Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/VI-nonabelian/](#).

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1. WHICH STEPS OF THE ABELIAN ARGUMENT USE COMMUTATIVITY

Recall the GL_1 chain of reasoning [5]. A KMS_β state is a measure ν on Y_S scaling by $N\mathfrak{a}^{-\beta}$; the symmetry group G_S acts on the simplex; a character $\chi \in \widehat{G_S}$ fails to be an obstruction exactly when $\chi \circ \sigma$ is a coboundary for π_β , which by Kakutani is the convergence of $\sum_{\mathfrak{p}} N\mathfrak{p}^{-\beta} |1 - \chi(\sigma_{\mathfrak{p}})|^2$; the obstruction characters form a *subgroup* $\Xi_\beta \leq \widehat{G_S}$; and the simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$.

Three of these steps are insensitive to commutativity and two are not.

- (i) The Kakutani step has two halves: a Dirichlet series in Frobenius data, and its identification with a coboundary condition for a cocycle on the tail relation. §2 shows that the series transfers, with the Hilbert–Schmidt norm in place of $|1 - z|$, and that the cocycle half transfers exactly when the Frobenius classes along S commute — the tail relation being generated by commuting shifts, a $U(n)$ -valued cocycle with prescribed non-commuting increments does not exist.
- (ii) The passage from convergence to a group structure uses that $\widehat{G_S}$ is a group under pointwise product. For non-abelian G the irreducibles are not closed under tensor product as a group law: $\rho_1 \otimes \rho_2$ decomposes. §3 replaces the subgroup by a Tannakian subcategory.
- (iii) The identification of the simplex as $\text{Prob}(G_S/\Xi_\beta^\perp)$ uses Pontryagin duality to convert a subgroup of the dual into a closed subgroup of the group. This is the step with no substitute at the level of duals; §3 supplies the substitute at the level of the group itself.
- (iv) The L -function dictionary is an Euler-product identity and transfers (§4), with different constants from those one might guess.
- (v) The construction of an algebra whose simplex realizes the answer does *not* transfer, and §6 says why.

Throughout, K is a number field, S a set of finite primes, and G a compact group equipped with a family of conjugacy classes $\text{Frob}_{\mathfrak{p}} \in G$ indexed by all but finitely many $\mathfrak{p} \in S$: typically G is a profinite quotient of $\text{Gal}(\bar{K}/K)$ unramified outside a finite set and $\text{Frob}_{\mathfrak{p}}$ the Frobenius class, but G may also be a Sato–Tate group and $\text{Frob}_{\mathfrak{p}}$ the Satake class, as in §5. All representations are continuous, unitary and finite-dimensional; sums over $\mathfrak{p} \in S$ omit the finitely many primes at which $\text{Frob}_{\mathfrak{p}}$ is undefined, which affects no convergence question. We write $\zeta_{K,S}(\beta) = \prod_{\mathfrak{p} \in S} (1 - N\mathfrak{p}^{-\beta})^{-1}$ as in [5], so that $\log \zeta_{K,S}(\beta) = \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} + O(1)$ for $\beta > 1/2$.

2. THE MATRIX KAKUTANI DICHOTOMY

Definition 2.1. For a representation $\rho : G \rightarrow U(n)$ put

$$(2.1) \quad \Xi_\beta(\rho, S) := \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} \|I_n - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2.$$

Lemma 2.2 (Trace form). *For $U \in U(n)$, $\|I_n - U\|_{\text{HS}}^2 = 2(n - \Re \text{Tr } U)$. Hence*

$$(2.2) \quad \Xi_\beta(\rho, S) = 2 \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} (n - \Re \text{Tr } \rho(\text{Frob}_{\mathfrak{p}})),$$

a sum of non-negative terms, each in $[0, 4n]$, depending only on the conjugacy class of $\rho(\text{Frob}_{\mathfrak{p}})$.

Proof. $\|I - U\|_{\text{HS}}^2 = \text{Tr}((I - U)^*(I - U)) = \text{Tr}(I - U - U^* + U^*U) = \text{Tr}(2I - U - U^*) = 2n - 2\Re \text{Tr } U$, using $U^*U = I$. Non-negativity and the bound follow from $|\text{Tr } U| \leq n$; conjugation invariance of the trace gives the last claim. $\square$

Theorem 2.3 (Matrix Kakutani). *Let $\pi_\beta = \bigotimes_{\mathfrak{p} \in S} \mu_{t_\mathfrak{p}}$ with $t_\mathfrak{p} = N\mathfrak{p}^{-\beta}$ as in [5, Lem. 2.10], on $V = \mathbb{N}^{(S)}$ with the tail relation $\mathcal{R}_S$, and put $U_\mathfrak{p} = \rho(\text{Frob}_\mathfrak{p}) \in U(n)$ (a representative of the class).*

- (1) *A $U(n)$ -valued cocycle c_ρ on $\mathcal{R}_S$ with $c_\rho(v + e_\mathfrak{p}, v) = U_\mathfrak{p}$ for all $\mathfrak{p}$ and v exists if and only if the $U_\mathfrak{p}$, $\mathfrak{p} \in S$, commute pairwise.*
- (2) *Suppose they do. Then there is a measurable $g : V \rightarrow U(n)$ with $g(v + e_\mathfrak{p}) = U_\mathfrak{p} g(v)$ a.e. for every $\mathfrak{p}$ — i.e. c_ρ is a coboundary — if and only if $\Xi_\beta(\rho, S) < \infty$.*

Proof. (1) The cocycle identity along the two paths from v to $v + e_\mathfrak{p} + e_\mathfrak{q}$ gives $c_\rho(v + e_\mathfrak{p} + e_\mathfrak{q}, v) = U_\mathfrak{q}U_\mathfrak{p} = U_\mathfrak{p}U_\mathfrak{q}$; conversely, for commuting $U_\mathfrak{p}$ the formula $c_\rho(v + a, v) = \prod_\mathfrak{p} U_\mathfrak{p}^{a_\mathfrak{p}}$ is well defined and is a cocycle.

(2) Commuting unitaries are simultaneously diagonalizable: some unitary W satisfies

$$WU_\mathfrak{p}W^* = \text{diag}(\chi_1(\mathfrak{p}), \dots, \chi_n(\mathfrak{p})) \quad \text{for every } \mathfrak{p}, \quad \chi_r(\mathfrak{p}) \in \mathbb{T}.$$

Since $\|I - U_\mathfrak{p}\|_{\text{HS}}^2 = \sum_r |1 - \chi_r(\mathfrak{p})|^2$,

$$\Xi_\beta(\rho, S) = \sum_{r=1}^n \sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} |1 - \chi_r(\mathfrak{p})|^2.$$

If g exists, put $h = Wg$, so that $h(v + e_\mathfrak{p}) = \text{diag}(\chi_r(\mathfrak{p})) h(v)$ and each entry satisfies $h_{rs}(v + e_\mathfrak{p}) = \chi_r(\mathfrak{p}) h_{rs}(v)$. For each r some entry h_{rs} is nonzero on a set of positive measure, $h(v)$ being unitary; by [5, Lem. 3.1] $|h_{rs}|$ is a.e. constant, hence a.e. nonzero, and [5, Prop. 3.3], whose proof uses only $|\chi_r(\mathfrak{p})| = 1$, gives the convergence of the r -th series. Conversely, if all n series converge, [5, Prop. 3.4] gives measurable $g_r : V \rightarrow \mathbb{T}$ with $g_r(v + e_\mathfrak{p}) = \chi_r(\mathfrak{p}) g_r(v)$, and $g = W^* \text{diag}(g_1, \dots, g_n) W$ is the required coboundary. $\square$

Remark 2.4 (The series is the primary object). Without commutation of the Frobenius classes there is no cocycle on $\mathcal{R}_S$ to be a coboundary, and the dynamical half of the Kakutani dichotomy has no content; a non-abelian dynamical model would need a non-abelian acting semigroup, and Paper V of the series shows what that costs. What survives, and what everything below uses, is the Dirichlet series $\Xi_\beta(\rho, S)$ itself, with the following formal properties.

Lemma 2.5 (Additivity and submultiplicativity). *For representations ρ, ρ' of dimensions n, n' :*

$$(2.3) \quad \|I_{n+n'} - (\rho \oplus \rho')(g)\|_{\text{HS}}^2 = \|I_n - \rho(g)\|_{\text{HS}}^2 + \|I_{n'} - \rho'(g)\|_{\text{HS}}^2,$$

$$(2.4) \quad \|I_{nn'} - (\rho \otimes \rho')(g)\|_{\text{HS}}^2 \leq 2n' \|I_n - \rho(g)\|_{\text{HS}}^2 + 2n \|I_{n'} - \rho'(g)\|_{\text{HS}}^2,$$

$$(2.5) \quad \|I_n - \rho^\vee(g)\|_{\text{HS}} = \|I_n - \rho(g)\|_{\text{HS}}.$$

Proof. (2.3) is block-diagonality of $I - (\rho \oplus \rho')$. For (2.4) write $I - A \otimes B = (I - A) \otimes I + A \otimes (I - B)$ and use $\|X \otimes I_{n'}\|_{\text{HS}}^2 = n' \|X\|_{\text{HS}}^2$, $\|A \otimes Y\|_{\text{HS}}^2 = n \|Y\|_{\text{HS}}^2$ for A unitary, together with $\|X + Y\|^2 \leq 2\|X\|^2 + 2\|Y\|^2$. (2.5) is $\rho^\vee(g) = \rho(g)$ and invariance of the Hilbert–Schmidt norm under entrywise conjugation. $\square$

Remark 2.6 (Numerical check). Lemma 2.2 and (2.4) were checked on Haar-random unitaries, 200 in each of the dimensions $n = 1, 2, 3, 5$ and 500 pairs with $(n, n') = (2, 3)$, with maximal error $3.6 \cdot 10^{-15}$ and no violation; a check of algebra, recorded because the constants of §4 depend on it.

3. THE OBSTRUCTION IS A QUOTIENT, NOT A DUAL SUBGROUP

Proposition 3.1 (Failure of the abelian formulation). *Let G be a non-abelian compact group. Then the set of irreducible representations $\widehat{G}$ carries no group structure for which $\rho \mapsto \Xi_\beta(\rho, S)$ is a homomorphism-like invariant: $\rho_1 \otimes \rho_2$ is in general reducible, so $\widehat{G}$ is a based set with a fusion rule and not a group, and the annihilator $\Xi_\beta^\perp$ of a subset of $\widehat{G}$ is undefined. In particular the statement “ $\text{KMS}_\beta \cong \text{Prob}(G/\Xi_\beta^\perp)$ ” has no meaning as written.*

Proof. Immediate from the definitions; the abelian case is exactly the case where every $\rho_1 \otimes \rho_2$ is again irreducible. $\square$

Theorem 3.2 (Tannakian obstruction). *Let $\mathcal{T}_\beta(S)$ be the full subcategory of $\text{Rep}(G)$ on the objects ρ with $\Xi_\beta(\rho, S) < \infty$. Then $\mathcal{T}_\beta(S)$ contains the trivial representation and is closed under direct sums, subobjects, quotients, tensor products and duals; it is therefore a Tannakian subcategory of $\text{Rep}(G)$, and*

$$\mathcal{T}_\beta(S) = \text{Rep}(G/N_\beta), \quad N_\beta := \bigcap_{\rho \in \mathcal{T}_\beta(S)} \ker \rho,$$

a closed normal subgroup of G . When G is abelian, $N_\beta = \Xi_\beta^\perp$ and the description agrees with [5, Thm. 3.6].

Proof. The trivial representation gives all summands zero. Closure under $\oplus$, $\otimes$ and duals is Lemma 2.5: for $\otimes$ the bound (2.4) converts two convergent series into a convergent one, and n, n' are fixed. For subobjects, if $\rho = \rho_1 \oplus \rho_2$ then (2.3) gives $\Xi_\beta(\rho_i, S) \leq \Xi_\beta(\rho, S)$, and by unitarity every subobject is a direct summand; quotients are duals of subobjects. Fullness holds by construction. A full subcategory of $\text{Rep}(G)$ closed under these operations and containing the trivial object is the representation category of a quotient of G , by Tannaka–Krein for compact groups: the matrix coefficients of its objects span a $*$ -subalgebra of the algebra of representative functions, closed under the coproduct, hence the representative functions of G/N with N the common kernel. Normality and closedness of N_β hold because each $\ker \rho$ is a closed normal subgroup.

For G abelian, $\mathcal{T}_\beta$ is generated by the characters in Ξ_β and $\bigcap_{\chi \in \Xi_\beta} \ker \chi = \Xi_\beta^\perp$. $\square$

Remark 3.3 (What the theorem does and does not assert). Theorem 3.2 is a statement about representations and Dirichlet series; it identifies the natural non-abelian analogue of Ξ_β as the quotient G/N_β . It does *not* assert that any C^* -dynamical system has $\text{Prob}(G/N_\beta)$ as its KMS_β simplex. That is the content of §6, and it is open.

We also restrict to finite-dimensional continuous unitary representations of a compact group. For a profinite G these are the Artin representations, with finite image. An ℓ -adic representation is not unitary and is not covered; the GL_2 case of §5 is handled through the Sato–Tate group, a compact Lie group, and its Satake classes.

4. L -FUNCTIONS, WITH THE CORRECT CONSTANTS

Theorem 4.1 (Euler dictionary). *For $\beta > 1/2$ and ρ of dimension n unramified outside S ,*

$$(4.1) \quad \Xi_\beta(\rho, S) = 2n \log \zeta_{K,S}(\beta) - 2 \Re \log L_S(\beta, \rho) + O(1),$$

where $L_S(s, \rho) = \prod_{\mathfrak{p} \in S} \det(I - \rho(\text{Frob}_{\mathfrak{p}}) \text{Np}^{-s})^{-1}$ and the implied constant depends only on n and β .

Proof. $\log L_S(s, \rho) = \sum_{\mathfrak{p}} \sum_{m \geq 1} \frac{1}{m} \text{Tr}(\rho(\text{Frob}_{\mathfrak{p}})^m) \text{Np}^{-ms}$, and the terms with $m \geq 2$ are bounded by $n \sum_{\mathfrak{p}} \sum_{m \geq 2} \text{Np}^{-m\beta} = O_n(1)$ for $\beta > 1/2$. Hence $\Re \log L_S(\beta, \rho) = \sum_{\mathfrak{p}} \text{Np}^{-\beta} \Re \text{Tr} \rho(\text{Frob}_{\mathfrak{p}}) + O(1)$, and similarly $\log \zeta_{K,S}(\beta) = \sum_{\mathfrak{p}} \text{Np}^{-\beta} + O(1)$. Substituting into (2.2) gives (4.1). $\square$

Remark 4.2 (Two constants that are easy to get wrong). The natural guess $\log \zeta_{K,S}(n\beta) - \Re \log L_S(\beta, \rho) = \Xi_\beta(\rho, S) + O(1)$ is incorrect in two ways. The zeta factor is evaluated at β , not at $n\beta$: the n in (2.2) enters as the *coefficient* $2n$ of a single Dirichlet series, coming from $\text{Tr } I_n = n$, not as a dilation of its argument. And both terms carry a factor 2, from $\|I - U\|_{\text{HS}}^2 = 2(n - \Re \text{Tr } U)$. For $n = 1$ the correct identity reduces to $\Xi_\beta = 2 \log \zeta_{K,S}(\beta) - 2 \Re \log L_S(\beta, \chi) + O(1)$, consistent with [5, Rem. 3.11].

Corollary 4.3 (The full prime set). *Let $S = P_K$ and ρ irreducible and nontrivial, and suppose $L(s, \rho)$ is holomorphic and nonzero at $s = 1$ — Hecke–Landau for Artin representations, Jacquet–Shalika [2] for the standard L -function of a cuspidal automorphic representation. Then $\Xi_\beta(\rho, P_K) = \infty$ for every $0 < \beta \leq 1$: the obstruction is absent, $N_\beta = G$, and the would-be simplex is a point.*

Proof. For $\beta > 1$, (4.1) gives $\Xi_\beta(\rho, P_K) = 2n \log \zeta_K(\beta) + O(1)$, since $\log L(\beta, \rho)$ stays bounded as $\beta \downarrow 1$ by the hypothesis on $L(1, \rho)$; so $\Xi_\beta(\rho, P_K) \rightarrow \infty$ as $\beta \downarrow 1$. As Ξ_β is non-increasing in β , being a Dirichlet series with non-negative coefficients, $\Xi_\beta = \infty$ for all $\beta \leq 1$. $\square$

The convergent case $\Xi_1(\rho, P_K) < \infty$ would require $\Re \log L(\beta, \rho)$ to diverge like $n \log \zeta_K(\beta)$, i.e. a pole of order n at $s = 1$, which happens only for ρ trivial.

Corollary 4.4 (Chebotarev criterion, Artin case). *Let ρ have finite image and let $D_\rho = \{\mathfrak{p} \in S : \rho(\text{Frob}_\mathfrak{p}) \neq I_n\}$. Then*

$$\Xi_\beta(\rho, S) < \infty \iff \sum_{\mathfrak{p} \in D_\rho} N\mathfrak{p}^{-\beta} < \infty.$$

Proof. The image is finite, so $\{\|I - \rho(g)\|_{\text{HS}}^2 : g \in G, \rho(g) \neq I\}$ is a finite set of strictly positive numbers, bounded above by $4n$ and below by some $c > 0$. Hence the summand in (2.1) is between $cN\mathfrak{p}^{-\beta}$ and $4nN\mathfrak{p}^{-\beta}$ on D_ρ and zero off it. $\square$

Remark 4.5. Corollary 4.4 is the exact analogue of [5, Thm. 3.9] and shows that the Chebotarev machinery of that paper — in particular the density computations and the block constructions of [6, §5] — transfers unchanged to Artin representations, with D_ρ in place of the detecting set of a character. *It fails for infinite image:* if $\rho(\text{Frob}_\mathfrak{p})$ can approach I_n , the indicator of D_ρ is not comparable to $\|I - \rho(\text{Frob}_\mathfrak{p})\|_{\text{HS}}^2$ and only the quadratic form is correct. The GL_2 case below is exactly of this kind.

5. GL_2 : SATO–TATE FORBIDS SYMMETRY BREAKING, ANOMALIES PERMIT IT

Let $E/\mathbb{Q}$ be a non-CM elliptic curve, a_p the trace of Frobenius, and $a_p = 2\sqrt{p} \cos \theta_p$ with $\theta_p \in [0, \pi]$. The ℓ -adic representation of E is not unitary and does not fit Definition 2.1 directly; what does is the Sato–Tate group $G = SU(2)$ (the non-CM case, by Serre’s open image theorem [3]) with the Satake classes $\text{Frob}_p = \text{diag}(e^{i\theta_p}, e^{-i\theta_p})$ and ρ its standard representation, to which §§2–4 apply verbatim. Then $n = 2$ and

$$(5.1) \quad \|I_2 - \rho(\text{Frob}_p)\|_{\text{HS}}^2 = 2(2 - 2 \cos \theta_p) = 4(1 - \cos \theta_p) = 8 \sin^2(\theta_p/2).$$

Theorem 5.1 (Sato–Tate forbids the obstruction). *Suppose the θ_p , $p \in S$, are equidistributed in $[0, \pi]$ for the Sato–Tate measure $\mu_{ST} = \frac{2}{\pi} \sin^2 \theta d\theta$ in the sense that $\sum_{p \in S, p \leq x} p^{-\beta} f(\theta_p) \sim (\int f d\mu_{ST}) \sum_{p \in S, p \leq x} p^{-\beta}$ for continuous f . Then*

$$\Xi_\beta(\rho, S) \sim 4 \sum_{p \in S} p^{-\beta},$$

which diverges whenever the partition function does. Hence $\rho \notin \mathcal{T}_\beta(S)$ and no symmetry breaking arises from Sato–Tate distributed classes, at any β with $\zeta_S(\beta) = \infty$.

Proof. By (5.1) the summand is $f(\theta_p)$ with $f(\theta) = 4(1 - \cos \theta)$, continuous and bounded on $[0, \pi]$. Since $\int_0^\pi \cos \theta \sin^2 \theta d\theta = 0$,

$$\int_0^\pi f d\mu_{ST} = \frac{8}{\pi} \int_0^\pi (1 - \cos \theta) \sin^2 \theta d\theta = 4,$$

and the hypothesis applied to f gives the stated asymptotic. $\square$

Remark 5.2 (Numerical check). $\int_0^\pi (2 - 2 \cos \theta) d\mu_{ST} = 2.0000000000$ by numerical quadrature on 2×10^6 nodes, matching the exact value 2; the factor 2 of Lemma 2.2 then gives the constant 4 in Theorem 5.1.

Theorem 5.3 (Anomalous sparse sets exist). *Assume Sato–Tate for E — a theorem for non-CM curves over $\mathbb{Q}$ [4] — in the form: for every $\delta > 0$ the set $\{p : \theta_p \leq \delta\}$ has positive natural density, hence $\sum_{p \leq x, \theta_p \leq \delta} p^{-1} \rightarrow \infty$. Then there is $S \subset P_{\mathbb{Q}}$ with*

$$\sum_{p \in S} p^{-1} = \infty \quad \text{and} \quad \Xi_1(\rho, S) < \infty.$$

Proof. Build $S = \bigsqcup_{j \geq 1} S_j$ greedily. Given $S_1, \dots, S_{j-1}$, the hypothesis with $\delta = 2^{-j}$ gives $\sum_{p \leq x, \theta_p \leq 2^{-j}} p^{-1} \rightarrow \infty$, so a finite set S_j of primes with $\theta_p \leq 2^{-j}$, disjoint from the previous blocks, may be chosen with $\sum_{p \in S_j} p^{-1} \geq 1$. Then $\sum_{p \in S} p^{-1} \geq \sum_j 1 = \infty$. On the other hand, by (5.1) and $\sin^2(\theta/2) \leq \theta^2/4$,

$$\Xi_1(\rho, S) = \sum_j \sum_{p \in S_j} p^{-1} \cdot 8 \sin^2(\theta_p/2) \leq \sum_j 2 \cdot 4^{-j} \sum_{p \in S_j} p^{-1}.$$

Choosing S_j additionally with $\sum_{p \in S_j} p^{-1} \leq 2$, which is possible since single terms p^{-1} can be made arbitrarily small, gives $\Xi_1(\rho, S) \leq \sum_j 4 \cdot 4^{-j} < \infty$. $\square$

Remark 5.4 (What has and has not been shown). Theorem 5.3 produces a set on which the analytic obstruction is finite: the Sato–Tate classes of a non-CM elliptic curve become “invisible” along a deliberately chosen set of primes with anomalously small angle. By Theorem 3.2 applied to $G = SU(2)$ it follows that $\rho \in \mathcal{T}_1(S)$ and hence $N_1 \subseteq \ker \rho = \{1\}$, so the Tannakian quotient $G/N_1 = SU(2)$ is the whole Sato–Tate group, nontrivial and non-abelian.

What has *not* been shown is that this residual symmetry acts on a simplex of KMS states of any algebra, for the reason of §6. The phrase “non-abelian symmetry breaking” is therefore premature: we have a non-abelian obstruction group, not yet a broken phase. The construction also assumes Sato–Tate as an equidistribution input; for non-CM elliptic curves over $\mathbb{Q}$ this is a theorem, but only the qualitative positive-density statement is used.

6. THE ALGEBRA IS NOT CONSTRUCTED

Remark 6.1 (Where the GL_1 proof stops). The classification in [5] runs: (a) KMS_β states of $C(Y_S) \rtimes J_S$ correspond bijectively to $\text{Na}^{-\beta}$ -scaling measures on Y_S , by the conditional expectation onto the diagonal; (b) such measures form a simplex on which G_S acts; (c) the ergodic decomposition of that action is computed by the Kakutani criterion. Steps (a) and (b) require the coefficient algebra to be commutative and the semigroup to be Ore, so that a canonical expectation exists and the crossed product is a groupoid algebra.

The candidate $\mathcal{A}_{K,S}^{GL_n} = C(Y_S^{(n)}) \rtimes M$ fails both at present. The multiplicative structure M is not a semigroup for the obvious choices: $M_n(J_S)$ is not closed under multiplication with the required cancellation, and GL_n over a semiring of ideals has no Ore condition available to us. And the Connes–Marcolli GL_2 system [1], which does exist, is not of this shape: its symmetry is the action of $GL_2(\mathbb{A}_f)$ by endomorphisms of a noncommutative space of two-dimensional lattices,

and its arithmetic is that of modular functions and the modular field, not of a Galois group acting through Frobenius classes. There is no non-abelian class field theory realized in it, which is why its KMS theory does not take the form $\text{Prob}(G/N)$.

The honest statement is therefore: §§2–4 determine what the answer would have to be, and identify the correct algebraic shape of the obstruction, but the system whose equilibrium states realize $\text{Prob}(G/N_\beta)$ is not available. Two routes seem worth trying. One is to look for a groupoid model directly: a groupoid whose unit space carries a G -action with the tail relation of [5] and whose isotropy realizes N_β ; this bypasses the semigroup difficulty entirely. The other is to stay with GL_1 but to enlarge the coefficient algebra, replacing $C(Y_S)$ by a noncommutative algebra carrying ρ ; here the Kakutani analysis of §2 applies verbatim, and the question is whether the resulting crossed product has a tractable expectation.

7. ASSESSMENT

Statement	Status	Reference
$\ I - U\ _{\text{HS}}^2 = 2(n - \Re \text{Tr } U)$	true	Lem. 2.2
Kakutani dichotomy for $U(n)$ -cocycles	iff the classes commute	Thm. 2.3
Ξ_β a subgroup of $\widehat{G}$	meaningless for non-abelian G	Prop. 3.1
$\text{Prob}(G/\Xi_\beta^\perp)$	meaningless as written	Prop. 3.1
$\mathcal{T}_\beta = \text{Rep}(G/N_\beta)$	true; the correct replacement	Thm. 3.2
$\log \zeta_{K,S}(n\beta) - \Re \log L_S = \Xi_\beta + O(1)$	false	Rem. 4.2
$\Xi_\beta = 2n \log \zeta_{K,S}(\beta) - 2\Re \log L_S + O(1)$	true	Thm. 4.1
$\Xi_\beta(\rho, P_K) = \infty$, $\rho \neq 1$ irreducible, $\beta \leq 1$	true, via $L(1, \rho) \neq 0, \infty$	Cor. 4.3
Chebotarev criterion via D_ρ	true for finite image only	Cor. 4.4
Sato–Tate $\Rightarrow$ symmetry breaking	false: forbids it	Thm. 5.1
anomalous sparse S with $\Xi_1 < \infty$	true, under Sato–Tate	Thm. 5.3
$\mathcal{A}_{K,S}^{GL_n}$ constructed	no	§6

The pattern is the one already familiar from Paper III of the series: the arithmetic content — a Dirichlet series in Frobenius data, an L -function on $\Re s = 1$ — generalizes with no difficulty, because none of it used commutativity. What used commutativity was, first, the dynamical half of the Kakutani step, which needs a cocycle on a relation generated by commuting shifts, and second the *duality* converting the series into a statement about a space of states; the substitute for the latter is Tannakian rather than Pontryagin, a quotient of the group rather than a subgroup of its dual, and for the former there is none. The missing ingredient is not a theorem but an object, namely an algebra whose equilibrium states are indexed by G/N_β .

Three questions. Is N_β the intersection of the kernels of *all* representations in $\mathcal{T}_\beta$, or is it already cut out by the irreducible ones of bounded dimension? For G with an open image in $GL_2(\widehat{\mathbb{Z}})$ this is a question about which quotients can be realized by anomalous prime sets. Second, Theorem 5.3 constructs S for a fixed ρ ; can one construct S realizing a *prescribed* quotient G/N , as [6, Thm. 5.1] does in the abelian case for prescribed $\bigoplus \mathbb{Z}/2\mathbb{Z}$? Third, and most important, is the groupoid model of Remark 6.1 available? Without it this paper describes an obstruction group with nothing yet to obstruct.

REFERENCES

- [1] A. Connes, M. Marcolli, *From physics to number theory via noncommutative geometry*, in: Frontiers in Number Theory, Physics, and Geometry I, Springer, 2006, 269–347.
- [2] H. Jacquet, J. A. Shalika, *A non-vanishing theorem for zeta functions of GL_n* , Invent. Math. 38 (1976/77), 1–16.

- [3] J.-P. Serre, *Propriétés galoisiennes des points d'ordre fini des courbes elliptiques*, Invent. Math. 15 (1972), 259–331.
- [4] R. Taylor, *Automorphy for some ℓ -adic lifts of automorphic mod ℓ Galois representations. II*, Publ. Math. IHES 108 (2008), 183–239.
- [5] R. Chen, *KMS state uniqueness and phase transitions in localized Bost–Connes systems*, preprint, 2026, <https://doi.org/10.5281/zenodo.22152101>.
- [6] R. Chen, *Localized Bost–Connes systems I: factor type and the spectrum of transitions*, preprint, 2026, DOI 10.5281/zenodo.22160827.

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